

Empirical Proof Record: VALIDATED

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Proof ID: 7ce1b27cb2b11e240c9bbd02cfb02f8e...

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1. Claim

Across a trajectory with spaced re-exposures, the substrate keeps recognising every re-exposed stimulus: (for every same-stimulus re-exposure pair (i, j) in the trajectory, $\text{Jaccard}(\text{concepts_i}, \text{concepts_j}) \geq 0.80$). The reported floor is the worst pair across the whole run.

- **Operationalisation:** Run a schedule of $\text{len}(\text{stimuli})$ unique stimuli each repeated $3\times$ and interleaved ($\text{gap} = \text{len}(\text{stimuli})$) through Kimera’s entity target (Takwin). For each stimulus, take the per-cycle “concepts” set at each exposure and compute Jaccard over all exposure pairs. $\text{recognition_jaccard_floor} = \min$ over all pairs across all stimuli; the LTL invariant holds iff $\text{floor} \geq \theta$.
- **Threshold:** $\text{recognition_jaccard_floor} \geq 0.8 \text{ jaccard}$
- **H0:** $\text{H0: floor} < 0.80$ — recognition collapses under manifold deformation somewhere in the run
- **H1:** $\text{H1: floor} \geq 0.80$ — recognition is stable across the whole trajectory (memory-as-deformation holds as a flow property)

2. Verdict

VALIDATED — observed 0.8636363636363636

recognition floor 0.8636 over 252 re-exposure pairs ($100 \text{ stimuli} \times 3 \text{ exposures}$); mean Jaccard 0.9936; 43 exposures produced no concept set; worst pair: stimulus 3 cycles $3 \leftrightarrow 103 = 0.8636$; control: same-stimulus median 1.000 vs cross-stimulus median 0.000, Mann-Whitney $p=1.5e-196$ (recognition IS significantly above the cross-stimulus baseline)

3. Evidence

Pillar	Statistic	Value	95% CI	Lib
0.flow.recognition_stability	recognition_jaccard_floor	0.8636363636363636	—	oph

4. Pre-registration

- Registered at: 2026-05-22T14:07:58.381470+00:00
- Config hash: f6bc53805d417e3efed14f11a888eb54...
- Data hash: 0337ab84541480cc72d72c54d196983b...

5. Signature

395baea0c4b982d438d5340d96670d88a7cbd4fa0468372fbddb24ae64b54c92